

LeetCode 684: Redundant Connection

Disjoint Set Union (DSU) Data Structure & Algorithm Notes

Document Information

Authored by: **Arpit Raj**

Institution: LNMIIT Jaipur

Date: August 3, 2026

C++ Solution Code

```
class dsu {
public:
    vector<int> parent;
    vector<int> size;

    dsu(int n) {
        parent.resize(n);
        size.resize(n, 1);
        for(int i = 0; i < n; i++) {
            parent[i] = i;
        }
    }

    int findparent(int node) {
        if(parent[node] == node) {
            return node;
        }
        parent[node] = findparent(parent[node]);
        return parent[node];
    }

    void merge(int u, int v) {
        int pu = findparent(u);
        int pv = findparent(v);

        if(pu == pv) return;

        if(size[pu] < size[pv]) {
            swap(pu, pv);
        }

        parent[pv] = parent[pu];
        size[pu] += size[pv];
    }
};

class Solution {
public:
    vector<int> findRedundantConnection(vector<vector<int>>& edges) {
        vector<int> ans;
        int n = edges.size();
        dsu ds(n + 1);

        for(auto &it : edges) {
```

```
int u = it[0];
int v = it[1];

if(ds.findparent(u) == ds.findparent(v)) {
    ans.clear();
    ans.push_back(u);
    ans.push_back(v);
}
else {
    ds.merge(u, v);
}

return ans;
}
};
```

© 2026 Arpit Raj. All rights reserved.